

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	G. Santosh Kumar	Reg. No.:	192524020
Section / Batch:	2025	Date:	31/08/2026

Code of Conduct

I G. Santosh Kumar _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Santosh Kumar _____

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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My Assessments

Course: CSA0305RA

PSEUDOCODE WRITING TASK

OPEN ✓ Completed

Review →

CO5_AT1_Pseudocode Writing Task

5/5 answered

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

OPEN ✓ Completed

Review →

DEBUGGING & OPTIMIZATION TASK

10/10 answered

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

OPEN ✓ Completed

Review →

DEBUGGING & OPTIMIZATION TASK

10/10 answered

CO3_AT3_MCQ

OPEN ✓ Completed

Review →

Multiple Choice Questions - Non Linear Data Structures

25/25 answered

CO1_AT1_C CODE OUTPUT PREDICTION

OPEN ✓ Completed

Review →

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PSEUDOCODE WRITING TASK

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QUESTIONS

Q1

Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Q2

Graph - Topological Sort

20 marks

Q3

Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's

20 marks

Q4

Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Q5

Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

5/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

a. All zones must be connected

b. Total installation cost must be minimized

c. No redundant connections (no cycles) allowed

d. The network must remain fully connected

Your Answer

INPUT: Graph G with V vertices and E edges with weights

OUTPUT: MST containing (V-1) edges with minimum total installation cost

BEGIN

Create an empty list MST_edges

total_cost = 0

Save Answer

Next →

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QUESTIONS

Q1 

Minimum Spanning Tree -
Kruskal's Algorithm

20 marks

Q2 

Graph - Topological Sort

20 marks

Q3

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```

Procedure SECOND_BEST_MST(V, E):

    mstEdges, mstCost <- KRUSKAL_MST(V, E)

    secondBestCost <- +INF
    secondBestEdges <- empty set

```

 Save Answer

